

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (CSE) (Credit System) Degree Examinations**

January 2023

20CS503 - OPERATING SYSTEMS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.***Unit – I**

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Briefly elaborate various operating system components. | 8 | L*2 | 1 | 1 |
| b) Describe the six services of operating system that are helpful for user. | 6 | L2 | 1 | 2 |
| c) List and explain with appropriate diagram the different states of a process. | 6 | L2 | 1 | 2 |
| 2. a) Briefly explain process scheduling with queueing diagram. | 6 | L2 | 1 | 2 |
| b) Explain the different IPC mechanism in detail. | 8 | L2 | 1 | 2 |
| c) Explain various multithreading models. | 6 | L2 | 1 | 1 |

Unit – II

3. a) Consider the following set of process.

Process	Arrival Time (msec)	Burst Time (msec)
P1	0	8
P2	1	4
P3	2	9
P4	3	6

- i) Draw the Gantt chart for preemptive shortest job first.
 ii) Compute the waiting time and turnaround time of each process for each of these algorithms.
 iii) Find the average turnaround time for each of these algorithms.
- | | | | | |
|---|---|----|---|---|
| b) Explain the Producer - Consumer problem. | 6 | L4 | 1 | 3 |
| c) What is Semaphore? Explain Implementation of semaphore with wait() and signal() operation. | 6 | L3 | 2 | 3 |
| | 8 | L2 | 2 | 3 |
4. a) Assume that there are 5 processes P0 through P4 and 3 types of resources. At time t_0 we have the following state as shown below:

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			

Apply Banker's Algorithm to answer the following.

- i) Is the system in a safe state?
 ii) If a request from a process P1(1,0,2) arrives, can it be granted?
- | | | | | |
|--|----|----|---|---|
| | 12 | L4 | 2 | 4 |
|--|----|----|---|---|
- b) What is race condition? Illustrate the race condition with the help of an example?
- | | | | | |
|--|---|----|---|---|
| | 8 | L2 | 2 | 1 |
|--|---|----|---|---|

Unit – III

5. a) Consider the page reference string for a memory with four frames. How many page faults will occur for FIFO, LRU and optimal page replacement algorithms? Which one of these three method is most efficient for the given reference string?
Reference string: 7,0,1,2,0,3,0,4,2,3,0,3,0,3,2,1,2,0,1,7,0,1. 10 L3 3 4
- b) What is page replacement? With the help of a diagram explain the concept of page replacement. 10 L2 3 2
6. a) With a neat diagram explain the concept of demand paging. 6 L3 3 4
- b) Define page fault. Explain how a page fault can be handled with a neat diagram. 8 L2 3 2
- c) Write a note on
i) Thrashing
ii) Allocation of Frames. 6 L2 3 2

Unit – IV

7. a) Discuss the different free space management approaches used to find the free blocks. 10 L2 4 1
- b) Explain the different ways in which computers can access the secondary storage. 10 L2 4 1
8. a) Explain the indexed file allocation methods with suitable diagrams. 8 L2 4 2
- b) Explain how the bad blocks in the hard disk is managed by operating system. 6 L2 4 1
- c) Explain the use of swap space management in operating system with an example. 6 L2 4 1

Unit – V

9. a) Explain different types of program threats. 6 L4 5 2
- b) Explain the various Ways of implementing a domain. 4 L2 5 1
- c) Explain the following.
i) Role based Access control
ii) Mandatory Access Control 10 L2 5 1
10. a) Explain the different categories of viruses. 10 L2 5 1
- b) Briefly explain the various types on breach of security with suitable example. 10 L2 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
